

```
#include <LiquidCrystal.h>
```

```
#include <stdio.h>
```

```
//LiquidCrystal lcd(6, 7, 5, 4, 3, 2);
```

```
LiquidCrystal lcd(13, 12, 14, 27, 26, 25);
```

```
unsigned char rcv,count,gchr,gchr1,robos='s';
```

```
//char pastnumber[11]="";
```

```
#include <WiFi.h>
```

```
#include <HTTPClient.h>
```

```
HTTPClient http;
```

```
const char *ssid = "iotserver";
```

```
const char *password = "iotserver123";
```

```
int httpResponseCode;
```

```
String servername = "http://projectsfactoryserver.in/storeddata.php?name=";
```

```
String accountname = "iot1720";
```

```
String field1 = "&s1=";
```

```
String field2 = "&s2=";
```

```
String field3 = "&s3=";
```

```
String field4 = "&s4=";
```

```
String field5 = "&s5=";
```

```
String payload="";
```

```
String ultra_string="";
```

```
String fire_string="";
```

```
String gas_string="";
```

```
int sti=0;
```

```
String inputString = "";    // a string to hold incoming data
```

```
boolean stringComplete = false; // whether the string is complete
```

```
int sti1=0;
```

```
String inputString1 = "";    // a string to hold incoming data
```

```
boolean stringComplete1 = false; // whether the string is complete
```

```
const int trigPin = 23;
```

```
const int echoPin = 22;
```

```
int m1a = 21;
```

```
int m1b = 19;
```

```
int m2a = 18;
```

```
int m2b = 5;
```

```
int fire = 2;
```

```
int gas = 4;
```

```
int buzzer = 2;
```

```
int rtr1=0;
```

```
int dist1=0,dist2=0,dist3,sts1=0,sts2=0;
```

```
long duration;
```

```
int distanceCm, distanceInch;
```

```
void iot_send()
```

```
{
```

```
    lcd.setCursor(15,1);lcd.write('U');
```

```
    http.begin(servername + accountname + field1 + ultra_string + field2 + fire_string + field3  
+ gas_string);
```

```
    httpResponseCode = http.GET();
```

```
    if(httpResponseCode>0)
```

```
    {
```

```
        payload="";
```

```
        //Serial.print("HTTP Response code: ");
```

```
        //Serial.println(httpResponseCode);
```

```
        payload = http.getString();
```

```
        //Serial.println(payload);
```

```
    }
```

```

else
{
;
//Serial.print("Error code: ");
//Serial.println(httpResponseCode);
}
delay(3000);
//lcd.setCursor(0,1);lcd.print("Emergency ");
lcd.setCursor(15,1);lcd.write(' ');
}
unsigned int ultra_dist()
{int ud=0;
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
duration = pulseIn(echoPin, HIGH);
distanceCm= duration*0.034/2;
ud = distanceCm;
return ud;
}

void setup()
{
Serial.begin(9600);//serialEvent();

pinMode(trigPin, OUTPUT); // Sets the trigPin as an Output

```

```
pinMode(echoPin, INPUT); // Sets the echoPin as an Input
```

```
pinMode(m1a, OUTPUT); pinMode(m1b, OUTPUT);
```

```
pinMode(m2a, OUTPUT); pinMode(m2b, OUTPUT);
```

```
pinMode(buzzer, OUTPUT);
```

```
pinMode(fire, INPUT);pinMode(gas, INPUT);
```

```
digitalWrite(m1a, LOW); digitalWrite(m1b, LOW);
```

```
digitalWrite(m2a, LOW); digitalWrite(m2b, LOW);
```

```
digitalWrite(buzzer, HIGH);
```

```
lcd.begin(16, 2);
```

```
lcd.print("IOT Industrial");
```

```
lcd.setCursor(0,1);
```

```
lcd.print("  Robot");
```

```
delay(2000);
```

```
WiFi.begin(ssid, password);
```

```
while(WiFi.status() != WL_CONNECTED)
```

```
{
```

```
    delay(500);
```

```
}
```

```
//Serial.println(WiFi.localIP());
```

```
delay(3000);
```

```
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("U:"); //2-3-4,0  
lcd.setCursor(5, 0);  
lcd.print("F:"); //7-8-9,0  
lcd.setCursor(10, 0);  
lcd.print("G:"); //12-13-14,0
```

```
//serialEvent();  
}
```

```
int dista=0;  
int cntlmk=0;  
void loop()  
{  
  dista=0;  
  dist1=0;  
  dist2=0;  
  
  for(rtr1=0;rtr1<5;rtr1++)  
  {  
    dista = ultra_dist();  
    dist1 = (dist1 + dista);  
    delay(10);  
  }
```

```
dist1 = (dist1/5);  
lcd.setCursor(2,0);convertl(dist1);  
ultra_string="";  
if(dist1 < 10)  
{  
    digitalWrite(m1a, LOW); digitalWrite(m1b, LOW);  
    digitalWrite(m2a, LOW); digitalWrite(m2b, LOW);  
    digitalWrite(buzzer, LOW);  
    ultra_string="Obstacle_U:" + String(dist1);  
    iot_send();  
}  
else  
{  
    digitalWrite(buzzer, HIGH);  
    ultra_string="U:" + String(dist1);  
}  
  
fire_string="";  
if(digitalRead(fire) == LOW)  
{  
    lcd.setCursor(7,0);lcd.print("ON ");  
    digitalWrite(m1a, LOW); digitalWrite(m1b, LOW);  
    digitalWrite(m2a, LOW); digitalWrite(m2b, LOW);  
    digitalWrite(buzzer, LOW);  
    fire_string="ON";iot_send();
```

```
    goto mn1;
}
if(digitalRead(fire) == HIGH)
{
    lcd.setCursor(7,0);lcd.print("OFF");
    fire_string="OFF";
}
mn1:
gas_string="";
if(digitalRead(gas) == LOW)
{
    lcd.setCursor(12,0);lcd.print("ON ");
    digitalWrite(m1a, LOW); digitalWrite(m1b, LOW);
    digitalWrite(m2a, LOW); digitalWrite(m2b, LOW);
    digitalWrite(buzzer, LOW);
    gas_string="ON";
    iot_send();
    goto mn2;
}
if(digitalRead(gas) == HIGH)
{
    lcd.setCursor(12,0);lcd.print("OFF");
    gas_string="OFF";
}
mn2:
if(dist1 > 10 && digitalRead(fire) == HIGH && digitalRead(gas) == HIGH)
```



```
{  
    digitalWrite(buzzer, HIGH);  
}
```

```
delay(1000);
```

```
//cntlmk++;
```

```
if(cntlmk >= 10)
```

```
{cntlmk=0;
```

```
    Serial.print("*Ultrasonic:");
```

```
    Serial.print(dist1);
```

```
if(digitalRead(fire) == LOW)
```

```
{
```

```
    Serial.print("_Fire_ON");
```

```
    goto mna;
```

```
}
```

```
if(digitalRead(fire) == HIGH)
```

```
{
```

```
    Serial.print("_Fire_OFF");
```

```
}
```

```
mna:
```

```
if(digitalRead(gas) == LOW)
```

```
{
```

```
    Serial.print("_Gas_ON");
```

```
    goto mnb;
```

```
}
```

```
if(digitalRead(gas) == HIGH)
```

```
{
```

```
    Serial.print("_Gas_OFF");
```

```
}
```

```
mnb:
```

```
    Serial.print("#");
```

```
}
```

```
while(Serial.available() < 0)
```

```
{
```

```
    char chrt = (char)Serial.read();
```

```
    if(chrt == '@')
```

```
    {
```

```
        sti=1;
```

```
    }
```

```
    if(sti == 1)
```

```
    {
```

```
        inputString += chrt;
```

```
    }
```

```
    if(chrt == '$')
```

```
    {sti=0;
```

```
        stringComplete = true;
```

```
    }
```

```
    if(chrt == '*')
```

```
    {
```

```
    sti1=1;
}
if(sti1 == 1)
{
    inputString1 += chrt;
}
if(chrt == '#')
{sti1=0;
    stringComplete1 = true;
}
}
```

```
if(stringComplete)
{
    if(inputString[1] == 'f')
    {
        digitalWrite(m1a, HIGH); digitalWrite(m1b, LOW);
        digitalWrite(m2a, HIGH); digitalWrite(m2b, LOW);
    }
    if(inputString[1] == 'b')
    {
        digitalWrite(m1a, LOW); digitalWrite(m1b, HIGH);
        digitalWrite(m2a, LOW); digitalWrite(m2b, HIGH);
    }
    if(inputString[1] == 'l')
```

```

{
    digitalWrite(m1a, HIGH); digitalWrite(m1b, LOW);
    digitalWrite(m2a, LOW); digitalWrite(m2b, HIGH);
}

if(inputString[1] == 'r')
{
    digitalWrite(m1a, LOW); digitalWrite(m1b, HIGH);
    digitalWrite(m2a, HIGH); digitalWrite(m2b, LOW);
}

if(inputString[1] == 's')
{
    digitalWrite(m1a, LOW); digitalWrite(m1b, LOW);
    digitalWrite(m2a, LOW); digitalWrite(m2b, LOW);
}

}

if(stringComplete1)
{
    lcd.setCursor(0,1);lcd.print("      ");
    lcd.setCursor(0,1);lcd.print(inputString1);
    inputString1 = "";
    stringComplete1 = false;
}
}

```

```
void serialEvent()
```

```
{
```

```
;
```

```
}
```

```
/*
```

```
    sensorValue = analogRead(analogInPin);
```

```
    sensorValue = (sensorValue/9.31);
```

```
    lcd.setCursor(1,1); //rc
```

```
    lcd.print(sensorValue);
```

```
    Serial.print(sensorValue);
```

```
*/
```

```
void converts(unsigned int value)
```

```
{
```

```
    unsigned int a,b,c,d,e,f,g,h;
```

```
    a=value/10000;
```

```
b=value%10000;
```

```
c=b/1000;
```

```
d=b%1000;
```

```
e=d/100;
```

```
f=d%100;
```

```
g=f/10;
```

```
h=f%10;
```

```
a=a|0x30;
```

```
c=c|0x30;
```

```
e=e|0x30;
```

```
g=g|0x30;
```

```
h=h|0x30;
```

```
Serial.write(a);
```

```
Serial.write(c);
```

```
Serial.write(e);
```

```
Serial.write(g);
```

```
Serial.write(h);
```

```
}
```

```
void convertl(unsigned int value)
```

```
{
```

```
    unsigned int a,b,c,d,e,f,g,h;
```

```
a=value/10000;  
b=value%10000;  
c=b/1000;  
d=b%1000;  
e=d/100;  
f=d%100;  
g=f/10;  
h=f%10;
```

```
a=a|0x30;  
c=c|0x30;  
e=e|0x30;  
g=g|0x30;  
h=h|0x30;
```

```
// lcd.write(a);  
// lcd.write(c);  
lcd.write(e);  
lcd.write(g);  
lcd.write(h);  
}
```